

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE PHYSICS

Paper 2

Friday 31 May 2024

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- a protractor
- the Physics Equations Sheet (enclosed).

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you worked out your answer.

Information

- The maximum mark for this paper is 90.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	



J U N 2 4 9 2 0 3 2 0 1

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ANSWER IN THE SPACES PROVIDED**



Answer **all** questions in the spaces provided.

0 1

There is a scientific theory that describes the universe beginning as a very large explosion.

0 1 . 1

How many years ago did the universe begin?

[1 mark]

Tick (✓) **one** box.

14 thousand years

☐

14 million years

☐

14 thousand million years

☐

0 1 . 2

Give the name of the theory that describes the universe beginning as a very large explosion.

[1 mark]

0 1 . 3

Scientists use telescopes to observe the light from distant galaxies.

What is the name of the observed increase in the wavelength of light from distant galaxies?

[1 mark]

Tick (✓) **one** box.

Blue shift

☐

Green shift

☐

Red shift

☐

Question 1 continues on the next page

Turn over ►

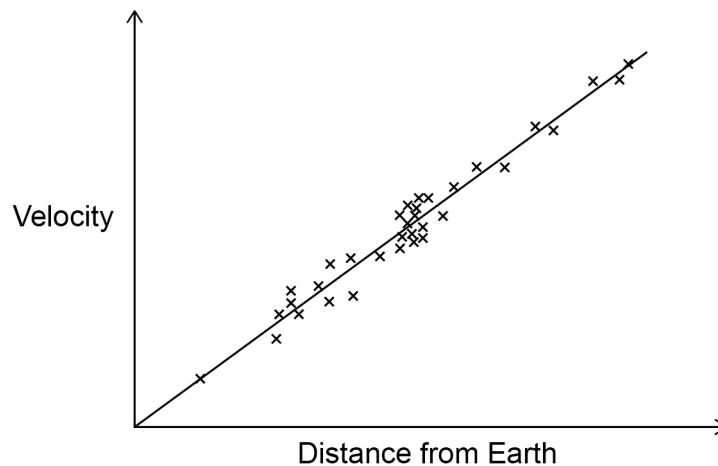


In 1929, a scientist called Edwin Hubble measured the wavelength of light from many galaxies.

He used these measurements to calculate how fast the galaxies were moving away from the Earth.

Figure 1 shows how the velocity of galaxies varies with distance from the Earth.

Figure 1



0 1 . 4 Which **two** conclusions were made from the graph?

[2 marks]

Tick (✓) **two** boxes.

The closer a galaxy is to the Earth, the longer the wavelength of light.

☐

The further away a galaxy is from the Earth, the faster the galaxy is moving.

☐

The galaxies are all slowing down due to gravity.

☐

The space that makes up the universe is expanding.

☐

The universe is made up of galaxies that contain hundreds of stars.

☐


0 1 . 5 There is a large amount of scatter about the line of best fit in **Figure 1**.

Suggest why there is a large uncertainty in the data in **Figure 1**.

[1 mark]

0 1 . 6 Using a different type of telescope, scientists have detected electromagnetic radiation that fills the universe.

This electromagnetic radiation is called CMBR.

What does CMBR stand for?

The last word has been completed for you.

[1 mark]

C –

M –

B –

R – Radiation

7

Turn over for the next question

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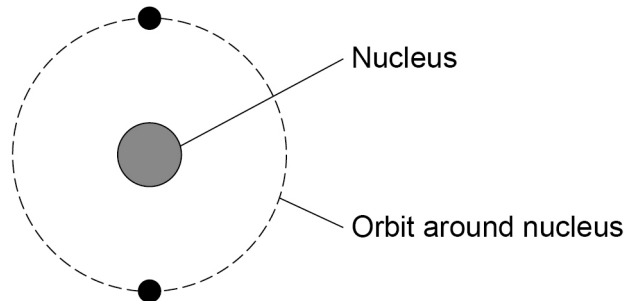


0	2
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Atoms are made up of electrons, neutrons and protons.

Figure 2 represents the structure of an atom.

Figure 2



0	2	.	1
---	---	---	---

Which **two** particles are located in the nucleus of the atom?

[1 mark]

1 _____

2 _____

0	2	.	2
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Which **two** particles in the atom have an electric charge?

[1 mark]

1 _____

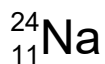
2 _____



The isotope sodium-24 is radioactive.

0 2 . 3 **Figure 3** represents an atom of sodium-24

Figure 3



How many electrons and how many neutrons are there in the atom of sodium in **Figure 3**?

[2 marks]

Electrons _____

Neutrons _____

0 2 . 4 Sodium-24 decays by emitting beta radiation.

What is the approximate range of beta radiation in air?

[1 mark]

Tick (✓) **one** box.

A few centimetres

☐

About one metre

☐

Over one kilometre

☐

Question 2 continues on the next page

Turn over ►



A doctor uses a sample of sodium-24 to monitor blood flow in the body of a patient.

0 2 . 5 The doctor needs to take safety precautions when working with radioactive samples.

Which **two** of the following safety precautions should the doctor take?

[2 marks]

Tick (✓) **two** boxes.

Limit exposure time

☐

Stand up at all times

☐

Tie long hair back

☐

Use heat-proof gloves

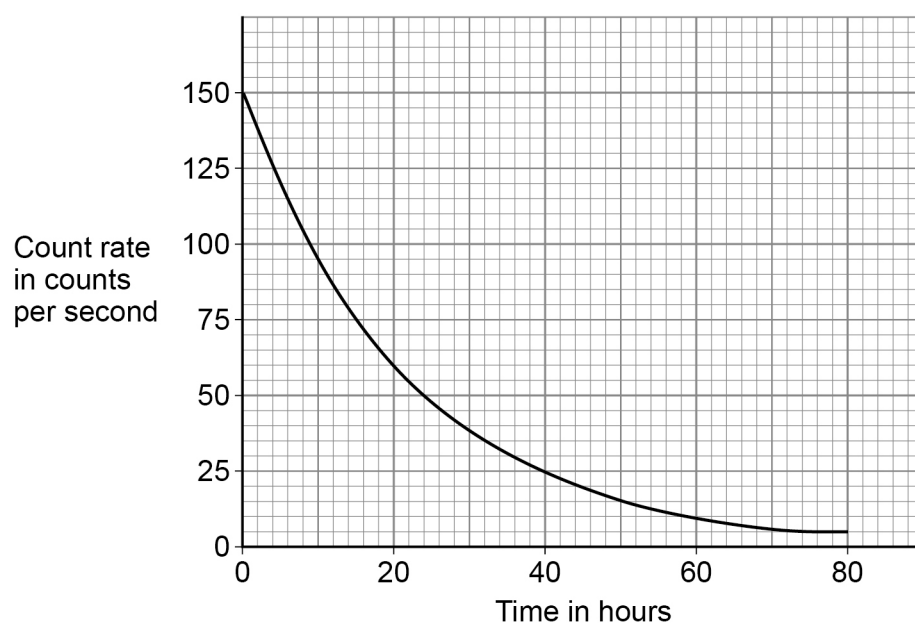
☐

Wear a lead-lined apron

☐

Figure 4 shows how the count rate from the sample of sodium-24 in the patient's blood changes with time.

Figure 4



0 2 . 6 Why does the count rate from the sample decrease as the sodium-24 decays?

[1 mark]

Tick (✓) **one** box.

The number of sodium-24 atoms decreases.

☐

The number of sodium-24 atoms stays the same.

☐

The number of sodium-24 atoms increases.

☐

0 2 . 7 **Figure 4** can be used to determine the half-life of sodium-24

What is meant by 'half-life'?

[1 mark]

0 2 . 8 After 4 half-lives, the count rate from the sample of sodium-24 will have dropped to a safe level.

Determine the time taken for the count rate from the sample to drop to a safe level.

Use **Figure 4**.

[3 marks]

Time taken = _____ hours

12

Turn over ►



0 3

Figure 5 shows a freezer that is used to keep food frozen.

Figure 5



Frozen food

The freezer is switched on continuously for 1 year.

0 3 . 1

How many hours are there in a year of 365 days?

[1 mark]

Tick (✓) **one** box.

24

☐

8760

☐

525 600

☐

31 536 000

☐


03.2 The freezer has a maximum power of 250 W.

The average power of the freezer is 20% of the maximum power.

Calculate the average power of the freezer.

Give your answer in kW.

[3 marks]

Average power = _____ kW

03.3 Calculate the cost of the mains electricity the freezer uses in 1 year.

cost per kilowatt-hour = \$0.40

Use your answers from Question **03.1** and Question **03.2**.

[2 marks]

Cost = \$ _____

03.4 Most electrical appliances are designed to be efficient.

Explain why using efficient electrical appliances saves money.

[2 marks]



0 4

Figure 6 shows an athlete holding a large exercise ball.

Figure 6



The athlete throws the ball to the ground.

0 4 . 1

The ball has a mass of 20 kg. The ball hits the ground with a kinetic energy of 640 J.

Calculate the speed of the ball when it hits the ground.

Use the Physics Equations Sheet.

[3 marks]

Speed = _____ m/s



0	4	.	2
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When the ball hits the ground, it stops moving.

Explain the energy transfers when the ball hits the ground.

[3 marks]

6

Turn over for the next question

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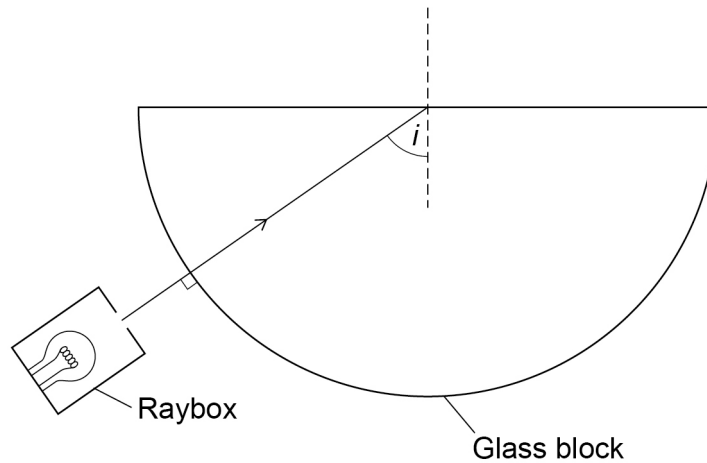


0 5

A student investigated the critical angle of light in a semi-circular glass block.

Figure 7 shows a ray of light entering a semi-circular glass block.

Figure 7



0 5 . 1

Describe how the student could accurately draw the path of the ray of light shown in **Figure 7**.

[3 marks]

0 5 . 2

Why is the ray of light **not** refracted when it enters the glass block?

[1 mark]

0 5 . 3

The angle of incidence (i) in **Figure 7** is greater than the critical angle.

Complete the path of the ray of light in **Figure 7**.

[2 marks]



Diamond has a refractive index of 2.42

0 5 . 4

A ray of light travels from air to diamond. The angle of incidence is 28.0°

Calculate the angle of refraction.

Give your answer to 3 significant figures.

[4 marks]

Angle of refraction (3 significant figures) = _____ $^\circ$

0 5 . 5

The refractive index of glass is 1.50

Explain why the critical angle for diamond is less than the critical angle for glass.

[3 marks]

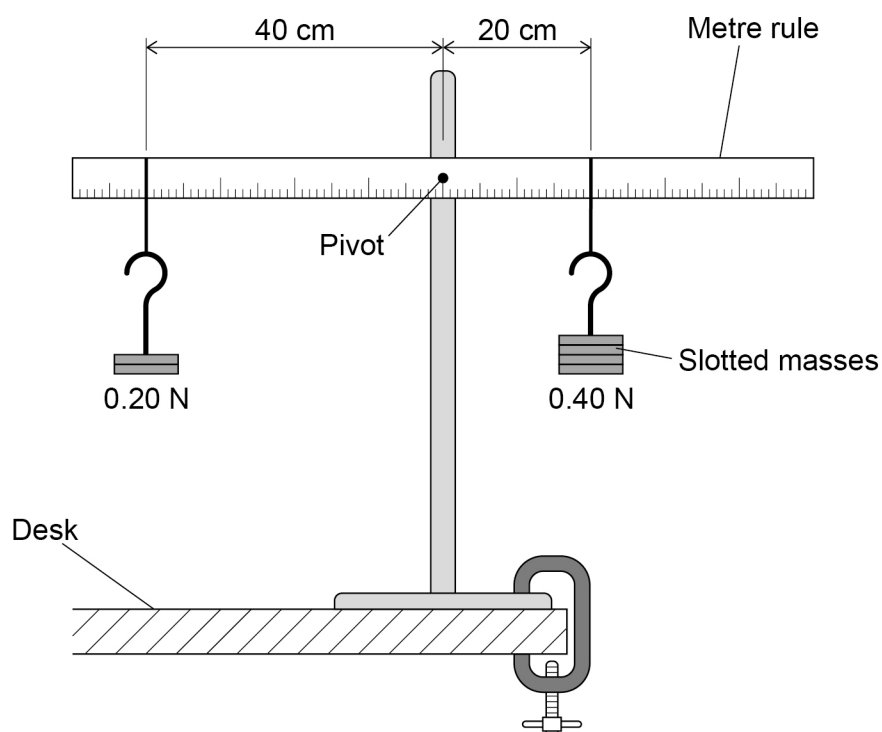


0 6

A student investigated moments.

Figure 8 shows the equipment the student used.

Figure 8



0 6 . 1

Explain **one** safety precaution the student used in the investigation.

Use **Figure 8**.

[2 marks]



0 6 . 2 Explain why the metre rule in **Figure 8** does **not** rotate.

Include a calculation in your answer.

[2 marks]

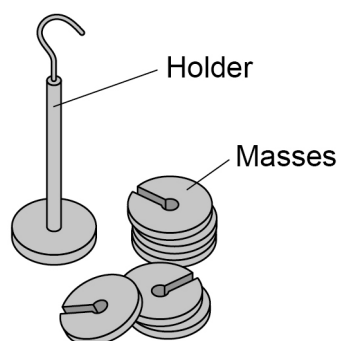
Question 6 continues on the next page

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0 6 . 3 **Figure 9** shows some different slotted masses and their holder.

Figure 9



Each slotted mass has a mass of 100 g. The holder also has a mass of 100 g.

A student placed some of the slotted masses on the holder.

The student held the slotted masses at a height of 0.80 m.

The masses and the holder had a total gravitational potential energy of 3.92 J.

gravitational field strength = 9.8 N/kg

Calculate the number of slotted masses that were on the holder.

Use the Physics Equations Sheet.

[4 marks]

Number of slotted masses = _____



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07

Figure 10 shows an electric car being driven at a constant speed.

Figure 10



07

1

The percentage of electric cars in the world is increasing.

Explain **one** advantage of using an electric car instead of a car that burns a fossil fuel.

[2 marks]



The car in **Figure 10** can be driven for a maximum time of 3.0 hours before the battery needs charging.

The battery stores 270 MJ of energy.

[6 marks]

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Potential difference = V

Question 7 continues on the next page

Turn over ►



0	7	.	3
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The maximum charge the battery can store depends on the physical size of the battery.

Describe **one** disadvantage of using a battery that can store a greater maximum charge.

[3 marks]

11



Turn over for the next question

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A student investigated the stretching of a spring.

The student wanted to show that the extension of the spring was directly proportional to the force applied to the spring.

You may include a diagram.

[6 marks]

[illegible]

Question 8 continues on the next page

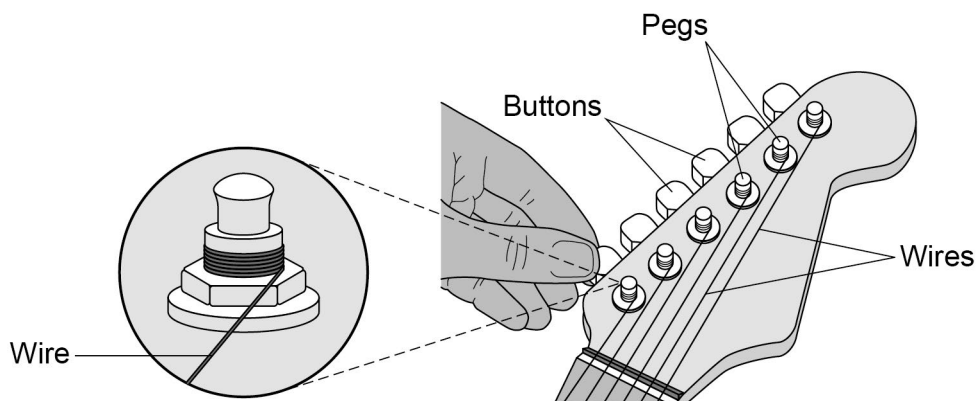
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A guitar is a musical instrument that contains strings made of metal wire.

Figure 11 shows a wire being adjusted by a guitar player.

Figure 11



When the button is turned, a peg turns. When the peg turns, a wire is stretched.

When the wire is stretched, it behaves like a spring.

0 8 . 2 The circumference of the peg is 0.020 m.

For each turn of the button, the wire extends by a distance equal to the circumference of the peg.

For 1.5 turns of the button, the elastic potential energy stored in the wire increases by 13.5 J.

Calculate the spring constant of the wire.

Give the unit.

[5 marks]

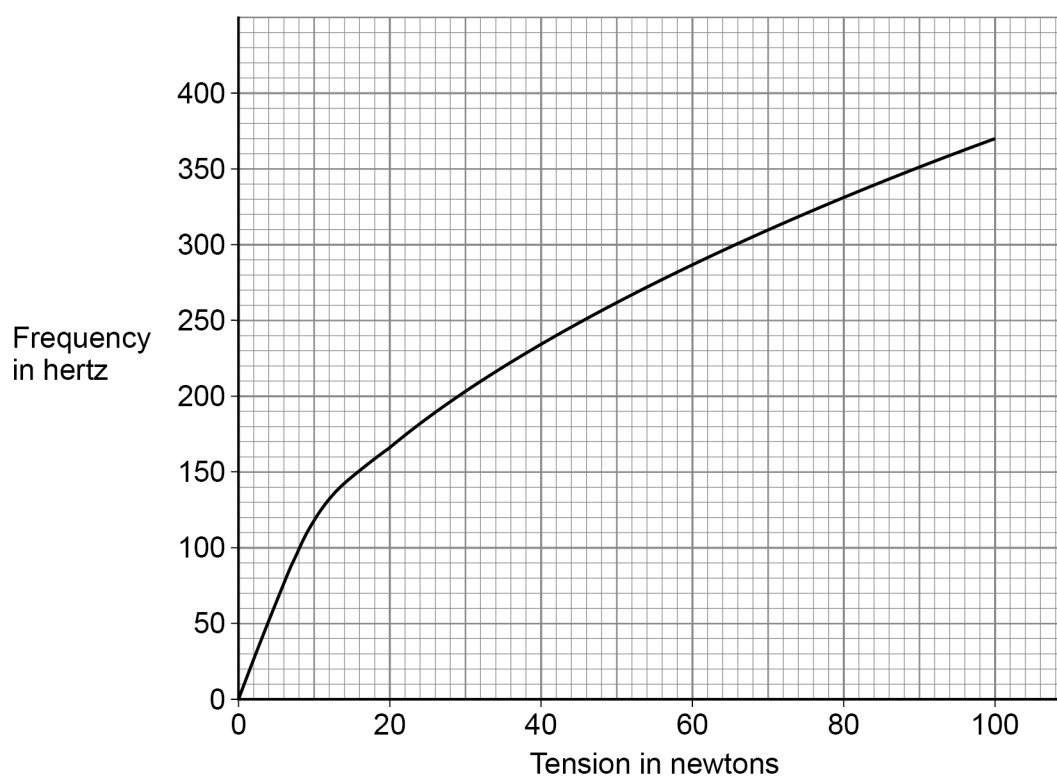
Spring constant = _____ Unit _____



0 8 . 3

Figure 12 shows how the frequency of a sound wave produced by the guitar wire is affected by the tension in the wire.

Figure 12



Explain how the frequency of the sound wave produced changes as the elastic potential energy stored in the wire increases.

[3 marks]



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In 1978, a probe entered the atmosphere of the planet Venus.

The probe fell through the atmosphere to the surface of Venus.

0	9	.	1
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The mass of the probe was 320 kg.

The weight of the probe on Venus was 2.88 kN.

Calculate the gravitational field strength of Venus.

[4 marks]

Gravitational field strength of Venus = _____ N/kg

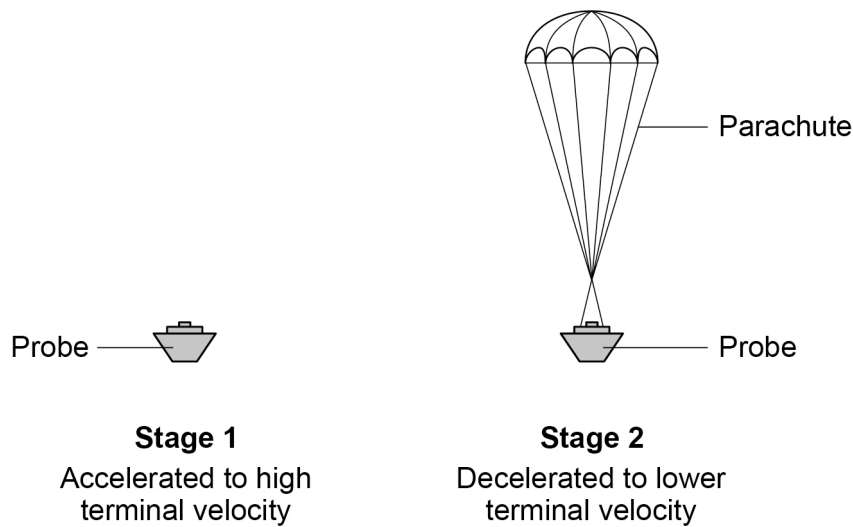
Question 9 continues on the next page

Turn over ►



Figure 13 shows the different stages of the probe's journey as it fell through the atmosphere.

Figure 13



In **stage 1**, the probe accelerated in the upper atmosphere of Venus until it reached a high terminal velocity.

In **stage 2**, a parachute then opened, and the probe fell at a new lower terminal velocity.

0 9 . 2 In **stage 1**, the probe accelerated until it reached terminal velocity.

Explain how the **forces** on the probe affected the acceleration of the probe during **stage 1**.

[3 marks]



0 9 . 3

In **stage 2**, a parachute opened.

When the parachute was opened, the **forces** on the probe caused the probe to decelerate.

Explain why.

[2 marks]

0 9 . 4

Stage 3 is **not** shown in **Figure 13**.

In **stage 3**, the parachute separated from the probe close to the surface of Venus.

The probe then fell without the parachute attached.

The probe reached terminal velocity.

The atmosphere of Venus makes the terminal velocity much less than it would be close to the surface of the Earth.

Explain why.

[2 marks]

11

END OF QUESTIONS



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